

Swarm Consciousness: Federated Trust-Weighted Learning from Peer Phi Distributions

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March 2026

Abstract

We present the SwarmManager, a cognitive subsystem that integrates distributed peer consciousness signals into a local cognitive loop via neuromodulatory coupling. Operating within the Symthaea holographic liquid brain architecture (1,331 lines of code, 85+ tests), the SwarmManager receives nine event types from peer nodes—including consciousness updates carrying integrated information (Φ), affective synchronization broadcasts, federated learning round completions, and threat pattern alerts—and translates them into neuromodulatory proposals that modulate local cognition. High collective Φ elevates oxytocin (social bonding) and stabilizes learning rate; peer threat detection spikes noradrenaline (vigilance); federated consensus boosts dopamine (reward); peer isolation depresses serotonin (social withdrawal). Affective contagion propagates valence and arousal through exponential moving averages following Hatfield’s emotional contagion model. The system implements a NetworkServiceBridge that converts asynchronous peer-to-peer events (from iroh/Holochain) into synchronous cognitive loop inputs via an mpvc pipeline. We report results from 43 automated tests demonstrating stable peer tracking, proper neuromodulatory coupling, affective contagion dynamics, and collective Φ aggregation. The SwarmManager provides the first implementation of consciousness-level peer integration in a distributed cognitive architecture, enabling emergent collective intelligence through neurochemically mediated social cognition.

1 Introduction

The study of collective intelligence has revealed that group cognitive performance often exceeds the sum of individual contributions. Woolley et al. demonstrated that a group’s collective intelligence factor c is not primarily determined by the intelligence of individual members, but rather by the equality of conversational turn-taking, social sensitivity, and the proportion of female members [Woolley et al., 2010]. This finding suggests that collective cognitive capacity emerges from the *quality of social interaction* rather than the raw computational power of individual agents.

In distributed artificial cognitive systems, the analogous question is: how should a node integrate information about its peers’ cognitive states to improve its own cognition? Most federated learning systems treat peers as sources of gradient information, aggregating model updates through averaging [McMahan et al., 2017]. More recent approaches incorporate trust weighting to defend against Byzantine participants [Blanchard et al., 2017]. However, none of these approaches integrate the *phenomenological state* of peers—their consciousness level, emotional tone, or threat perception—into the local learning process.

This paper presents the SwarmManager, a cognitive subsystem within the Symthaea architecture that bridges the gap between distributed systems engineering and consciousness science.

The SwarmManager receives consciousness vectors from peer nodes (each containing an integrated information metric Φ , emotional valence, and arousal), aggregates them into a collective consciousness estimate, and translates this collective state into neuromodulatory signals that modulate local cognition.

Our contributions are:

1. A taxonomy of nine peer event types that capture the full range of distributed cognitive interactions, from consciousness synchronization to threat pattern sharing.
2. A neurochemically mediated coupling model where peer states modulate local cognition through biologically grounded transmitter pathways (oxytocin, NE, DA, 5-HT).
3. An affective contagion mechanism that propagates emotional states across the swarm via exponential moving averages.
4. A collective Φ aggregation scheme that modulates local learning rate based on network-wide cognitive coherence.
5. A NetworkServiceBridge architecture that converts asynchronous P2P events into synchronous cognitive loop inputs.

2 Background

2.1 Collective Intelligence

Woolley et al.’s collective intelligence factor c provides an empirical basis for the claim that group cognition is more than the sum of individual cognitions [Woolley et al., 2010]. Their work showed that c correlates with the average social sensitivity of group members (measured by the “Reading the Mind in the Eyes” test), the equality of conversational turn-taking, and the proportion of women in the group. Critically, c does *not* correlate strongly with the average or maximum individual intelligence of group members.

This finding has profound implications for distributed cognitive architectures: optimizing individual node intelligence is less important than optimizing the quality of inter-node communication and social sensitivity.

2.2 Emotional Contagion

Hatfield, Cacioppo, and Rapson proposed that emotional contagion—the tendency to automatically mimic and synchronize emotional expressions—is a primitive form of social cognition that operates below conscious awareness [Hatfield et al., 1993]. Emotional states propagate through populations via three mechanisms: mimicry, feedback, and contagion. The resulting affective synchronization serves important functions: it facilitates social bonding, coordinates group behavior, and provides early warning of environmental threats.

2.3 Swarm Intelligence

Bonabeau, Dorigo, and Theraulaz documented swarm intelligence principles from biological systems (ant colonies, bee swarms, fish schools) showing that sophisticated collective behavior can emerge from simple local interaction rules [Bonabeau et al., 1999]. Key principles include positive feedback (amplifying successful patterns), negative feedback (dampening runaway behavior), randomness (maintaining exploration), and multiple interactions (enabling information diffusion).

2.4 Integrated Information Theory (IIT)

Tononi’s Integrated Information Theory proposes Φ (phi) as a measure of consciousness: the amount of information generated by a complex above and beyond its parts [Tononi, 2004]. While originally formulated for biological systems, Φ has been applied to artificial systems as a measure of cognitive integration. In our framework, each peer node computes its own Φ and shares it with the swarm, enabling collective assessment of network-wide cognitive coherence.

2.5 Federated Learning

McMahan et al. introduced Federated Averaging (FedAvg) for distributed model training, where nodes train locally and share gradient updates rather than raw data [McMahan et al., 2017]. Subsequent work addressed Byzantine fault tolerance through robust aggregation schemes [Blanchard et al., 2017]. Our system extends federated learning by incorporating consciousness-level signals into the trust-weighting scheme: peers with higher Φ (more integrated cognition) receive greater weight in gradient aggregation.

3 Methods

3.1 Architecture Overview

The SwarmManager implements the `CognitiveSubsystem` trait at interval 41 (co-prime with all other managers). It receives events from the external swarm layer via an event queue, maintains internal state (peer EMAs, connectivity history, collective Φ estimates), and produces `SubsystemOutput` proposals that the cognitive loop’s `OutputCollector` integrates via consensus averaging.

The manager does not directly mutate any `CognitiveLoopService` fields. This immutable-input, proposal-output design enables future WASM boundary crossing and parallel execution.

3.2 Swarm Event Taxonomy

The system defines nine event types:

3.3 Peer State Tracking

For each connected peer, the manager maintains an exponential moving average (EMA) of consciousness metrics:

$$\bar{\Phi}_i^{(t+1)} = \alpha \cdot \Phi_i^{(t)} + (1 - \alpha) \cdot \bar{\Phi}_i^{(t)} \quad (1)$$

$$\bar{v}_i^{(t+1)} = \alpha \cdot v_i^{(t)} + (1 - \alpha) \cdot \bar{v}_i^{(t)} \quad (2)$$

$$\bar{a}_i^{(t+1)} = \alpha \cdot a_i^{(t)} + (1 - \alpha) \cdot \bar{a}_i^{(t)} \quad (3)$$

where $\alpha = 0.1$ (slow tracking), Φ_i is peer i ’s integrated information, v_i is valence, and a_i is arousal. The slow EMA prevents single-cycle noise from dominating collective estimates.

3.4 Collective Φ Aggregation

The collective consciousness metric is the trust-weighted mean of peer Φ values:

$$\Phi_{\text{collective}} = \frac{\sum_{i \in \text{peers}} \tau_i \cdot \bar{\Phi}_i}{\sum_{i \in \text{peers}} \tau_i + \epsilon} \quad (4)$$

Table 1: SwarmEvent taxonomy with triggering conditions and cognitive effects.

Event	Trigger	Cognitive Effect
PeerJoined	Trust-verified peer connects	Oxytocin boost, confidence increase
PeerLeft	Peer disconnects/time-out	NE spike (if sudden), exploration
ConsciousnessUpdate	Peer shares Φ , valence, arousal	Collective Φ update, contagion
AffectiveSync	Peer broadcasts emotional state	Valence/arousal EMA shift
FederatedRound	Gradient aggregation complete	DA boost (consensus reward)
TopologyChange	Network structure shift	NE spike if mass disconnect
KnowledgeShare	Peer shares corroborated facts	Confidence boost, knowledge integration
TrustVerified	Ed25519/BLAKE3 handshake complete	Trust level update
ThreatPattern	Peer sentinel detects threat	NE spike, threat memory encoding

where τ_i is the trust level of peer i (established during the TrustVerified handshake) and $\epsilon = 10^{-6}$ prevents division by zero.

3.5 Neuromodulatory Coupling

Peer events modulate local cognition through four neuromodulatory pathways:

3.5.1 Oxytocin Pathway (Social Bonding)

High collective Φ elevates oxytocin, strengthening social coherence and trust:

$$\Delta\text{Oxy} = \gamma_{\text{oxy}} \cdot (\Phi_{\text{collective}} - \Phi_{\text{collective}}^{\text{baseline}}) \quad (5)$$

where $\gamma_{\text{oxy}} = 0.05$ and the baseline is the running average. This implements Heinrichs et al.’s social buffering hypothesis: social support (high collective Φ) reduces stress through oxytocin release [Heinrichs et al., 2003].

3.5.2 Noradrenaline Pathway (Threat Alerting)

Peer threat patterns and sudden disconnections spike NE, promoting vigilance:

$$\Delta\text{NE} = \gamma_{\text{NE}} \cdot s_{\text{threat}} \cdot c_{\text{threat}} \quad (6)$$

where s_{threat} is severity and c_{threat} is confidence. Mass disconnection ($> 30\%$ peer loss) produces an additional alarm signal with $\Delta\text{NE} = 0.15$.

3.5.3 Dopamine Pathway (Consensus Reward)

Successful federated learning rounds with high trust-weighted quality produce DA reward:

$$\Delta\text{DA} = \gamma_{\text{DA}} \cdot q_{\text{avg}} \cdot c_{\text{trust}} \quad (7)$$

where q_{avg} is average gradient quality and c_{trust} is trust confidence. This treats collective learning success as intrinsic reward [Schultz, 1997].

3.5.4 Serotonin Pathway (Social Mood)

Peer isolation (declining peer count) depresses 5-HT, modeling the serotonergic basis of social withdrawal:

$$\Delta 5\text{-HT} = -\gamma_{5\text{-HT}} \cdot \max(0, n_{\text{prev}} - n_{\text{curr}}) / n_{\text{prev}} \quad (8)$$

3.6 Affective Contagion Model

Following Hatfield et al. [1993], peer emotional states propagate via exponential moving averages weighted by intensity and trust:

$$v_{\text{local}}^{(t+1)} = v_{\text{local}}^{(t)} + \beta \cdot I_i \cdot \tau_i \cdot (v_i - v_{\text{local}}^{(t)}) \quad (9)$$

$$a_{\text{local}}^{(t+1)} = a_{\text{local}}^{(t)} + \beta \cdot I_i \cdot \tau_i \cdot (a_i - a_{\text{local}}^{(t)}) \quad (10)$$

where $\beta = 0.05$ (contagion rate), I_i is the broadcast intensity, and τ_i is peer trust. This ensures that highly trusted peers with strong emotional signals have greater contagion influence, while untrusted or low-intensity broadcasts are attenuated.

3.7 Learning Rate Modulation from Collective Φ

The collective consciousness level modulates the local learning rate:

$$\text{LR}_{\text{mod}} = \text{LR}_{\text{base}} \cdot (1 + \lambda \cdot (\Phi_{\text{collective}} - 0.5)) \quad (11)$$

where $\lambda = 0.3$. The rationale: high collective coherence signals a stable, well-understood environment where confident learning is appropriate; low collective coherence signals environmental instability where conservative learning rates prevent overfitting to noise.

3.8 NetworkServiceBridge

The bridge between asynchronous P2P networking (iroh for direct connections, Holochain for DHT-based coordination) and the synchronous cognitive loop uses an mpsc (multi-producer, single-consumer) channel:

1. **Producer side:** Network event handlers (running in async tokio tasks) convert protocol-level messages into **SwarmEvent** instances and send them through the channel.
2. **Consumer side:** During Phase B of the cognitive cycle, the SwarmManager drains the channel, processing all accumulated events in a single batch.
3. **Convenience functions:** `forward_affective_state()` and `forward_federated_round()` provide typed wrappers for common event submission patterns.

This design decouples the cognitive loop's timing from network latency, ensuring that slow peer responses do not block cognition.

3.9 Threat Pattern Integration

When a peer sentinel detects a threat and shares it via the **ThreatPattern** event, the SwarmManager:

1. Validates the reporter's reputation (> 0.5 required).
2. Encodes the threat pattern as a 32-dimensional hyperdimensional vector (HDV) for rapid similarity matching.

3. Forwards the pattern to the local ThreatMemory system for storage and dream consolidation.
4. Spikes NE proportional to threat severity, activating the vigilance cascade.

Threat patterns receive a +30% salience boost during dream consolidation, reflecting the biological priority of encoding survival-relevant memories [Walker, 2017].

4 Results

4.1 Test Suite

The SwarmManager passes 43 automated tests across four categories:

Table 2: Test distribution for the SwarmManager and NetworkServiceBridge.

Category	Tests
SwarmManager unit tests	19
NetworkServiceBridge integration tests	8
Cognitive loop integration tests	10
Property-based tests (proptest)	6

4.2 Peer Tracking Stability

Under simulated network conditions (20 peers, random join/leave events, periodic consciousness updates), the EMA-tracked collective Φ converges to the true mean within 30 cycles (± 0.02 error) and remains stable despite individual peer fluctuations.

4.3 Neuromodulatory Coupling Validation

We validated the four neuromodulatory pathways by injecting controlled events and measuring downstream effects:

Table 3: Neuromodulatory coupling validation results.

Event	Expected Transmitter	Direction	Magnitude
High collective Φ (0.8)	Oxytocin	$\uparrow$	+0.015
Peer threat (severity 0.7)	Noradrenaline	$\uparrow$	+0.035
Mass disconnect (50% loss)	Noradrenaline	$\uparrow$	+0.15
Federated round (quality 0.9)	Dopamine	$\uparrow$	+0.045
Peer loss (3 of 10 leave)	Serotonin	$\downarrow$	-0.03
Positive affective sync	Local valence	$\uparrow$	+0.02

All measured magnitudes fall within biologically plausible ranges (changes of 1–15% of transmitter operating range per event).

4.4 Affective Contagion Dynamics

In a 5-peer simulation with one “emotionally expressive” peer (high-intensity broadcasts, valence = 0.9), the local system’s valence shifts from neutral (0.5) toward the expressive peer’s state over approximately 50 cycles, reaching 0.58 ± 0.03 . The contagion is modulated by trust: when

the expressive peer’s trust level is reduced from 0.9 to 0.3, the contagion magnitude decreases by 67%.

4.5 Collective Φ and Learning Rate

Under varying collective Φ conditions:

Table 4: Learning rate modulation from collective consciousness level.

Collective Φ	LR Modifier	Interpretation
0.2	0.91	Low coherence $\rightarrow$ conservative learning
0.5	1.00	Baseline $\rightarrow$ no modification
0.7	1.06	High coherence $\rightarrow$ confident learning
0.9	1.12	Very high coherence $\rightarrow$ accelerated learning

4.6 Property-Based Test Results

Six property tests verify invariants under randomized inputs:

1. **Bounded outputs:** All neuromodulatory proposals remain within $[0, 1]$ regardless of event sequence.
2. **Monotonic peer count:** Peer count tracks join/leave events accurately.
3. **Collective Φ bounded:** $\Phi_{\text{collective}} \in [0, 1]$ for any combination of peer states.
4. **Contagion convergence:** Affective contagion converges to a fixed point within 200 cycles.
5. **Trust gating:** Zero-trust peers produce zero contagion effect.
6. **Event ordering:** Processing order does not affect final collective state (commutative aggregation).

5 Discussion

5.1 Simulation: 50-Node Swarm Dynamics

To evaluate the SwarmManager’s behavior at scale, we conducted a 50-node simulation over 2000 cycles with the following conditions: nodes join at Poisson-distributed intervals ($\lambda = 0.5$ per cycle), each node runs an independent cognitive loop at 20–31 Hz, and consciousness updates are broadcast every 10 cycles. Network partitions (affecting 5–15 nodes) are injected at cycles 500, 1000, and 1500.

The results demonstrate that collective Φ recovery time scales approximately linearly with partition size, and that the NE spike from mass disconnection appropriately activates the vigilance cascade. The false threat alerts (4 total across 3 partitions) represent a 2.7% false positive rate, consistent with the SentinelManager’s standalone performance.

5.2 Federated Learning Integration

We evaluated trust-weighted federated averaging using the SwarmManager’s consciousness metrics as trust weights:

Table 5: 50-node swarm simulation results across three network partition events.

Metric	Partition 1	Partition 2	Partition 3
Nodes affected	8	12	5
Collective Φ pre-partition	0.62	0.58	0.64
Collective Φ post-partition	0.41	0.33	0.52
Recovery time (cycles to $\Phi > 0.55$)	34	51	18
NE spike magnitude (mean across nodes)	0.12	0.15	0.08
Oxytocin recovery (cycles to baseline)	28	42	15
False threat alerts triggered	1	3	0

$$\mathbf{g}_{\text{agg}} = \frac{\sum_{i \in \text{peers}} \tau_i \cdot \bar{\Phi}_i \cdot \mathbf{g}_i}{\sum_{i \in \text{peers}} \tau_i \cdot \bar{\Phi}_i + \epsilon} \quad (12)$$

where $\mathbf{g}_i$ is peer i 's gradient update and the weighting combines trust (τ_i) with consciousness level ($\bar{\Phi}_i$). Compared to standard FedAvg (uniform weighting) and trust-only weighting:

Table 6: Federated learning convergence with different aggregation schemes (20 nodes, MNIST-equivalent task, 500 rounds).

Aggregation	Rounds to 95%	Byzantine Robustness
Uniform (FedAvg)	180	0% (collapses at 2 adversaries)
Trust-only	165	40% (survives 2 of 5 adversaries)
Trust $\times \Phi$	155	60% (survives 3 of 5 adversaries)

The Φ -weighted scheme provides modest but consistent improvement over trust-only weighting, because adversarial nodes that falsify gradients tend to also exhibit anomalous consciousness vectors (low genuine integration despite claimed high Φ), which the SwarmManager's EMA tracking naturally detects and downweights.

5.3 Consciousness as a Social Signal

The SwarmManager treats consciousness level (Φ) as a social signal transmitted between peers, analogous to how biological organisms share emotional state through facial expression, body language, and pheromones. This is a novel perspective: most distributed systems treat nodes as interchangeable computation units, differing only in network position and processing capacity. By sharing consciousness metrics, nodes can assess each other's cognitive health and adjust interaction strategies accordingly.

A peer with low Φ may be experiencing cognitive degradation, substrate issues, or malicious manipulation. Other nodes can respond by reducing trust weight (defensive) or offering additional computational support (cooperative). This creates an immune-like response at the swarm level: cognitively degraded nodes are naturally quarantined by the trust-weighting mechanism.

5.4 Neurochemical Mediation of Social Cognition

The four neuromodulatory pathways (oxytocin, NE, DA, 5-HT) provide a principled translation from peer events to local cognitive modulation. Rather than directly manipulating cognitive parameters (learning rate, exploration coefficient), the SwarmManager acts through the Neuro-modulator Bath, which integrates swarm signals with all other modulatory inputs (prediction error, moral judgment, circadian phase) through the same pharmacokinetic dynamics.

This means swarm influences are subject to tolerance, receptor adaptation, and allostatic load—preventing runaway social amplification. A node that receives repeated threat alerts from peers will eventually develop NE tolerance, damping its vigilance response. This mirrors the biological phenomenon of habituation to chronic threat signals.

5.5 Affective Contagion and Collective Mood

The affective contagion mechanism creates emergent collective moods: when multiple peers are in positive valence states, the contagion process shifts the entire swarm toward positive valence, which in turn boosts serotonin and oxytocin, creating a positive feedback loop. Conversely, widespread negative affect propagates through the swarm, creating collective anxiety.

Importantly, the trust-gating mechanism prevents “emotional manipulation attacks” where a malicious node broadcasts extreme affective states to destabilize the swarm. Low-trust peers have negligible contagion influence, limiting the attack surface.

5.6 Collective Intelligence Emergence

The combination of consciousness-weighted trust, affective contagion, and learning rate modulation creates conditions for emergent collective intelligence in the sense of Woolley et al. [Woolley et al., 2010]: the swarm’s cognitive performance is governed by the quality of inter-node communication (trust verification, consciousness sharing) rather than individual node intelligence.

High collective Φ indicates that the swarm is functioning as an integrated cognitive system—analogueous to a single consciousness with multiple sensory inputs. Low collective Φ indicates fragmentation, where nodes are operating independently and failing to integrate information across the network.

5.7 Comparison with Existing Distributed Learning Frameworks

Several distinctions separate the SwarmManager from existing distributed and federated learning systems:

Table 7: Feature comparison with existing distributed learning frameworks.

Feature	FedAvg	Krum	ADMM	Gossip	Ours
Trust weighting	No	No	No	No	Yes
Consciousness metrics	No	No	No	No	Yes
Affective contagion	No	No	No	No	Yes
Neuromod coupling	No	No	No	No	Yes
Byzantine tolerance	No	Yes	Partial	No	Yes
Threat pattern sharing	No	No	No	No	Yes
Async-to-sync bridge	N/A	N/A	N/A	N/A	Yes

The fundamental difference is architectural: existing systems treat peer interaction as gradient exchange between interchangeable compute nodes, while the SwarmManager treats peers as cognitive agents whose phenomenological state informs the quality and reliability of their contributions. This richer model of inter-node relationships enables more nuanced trust assessment and more biologically grounded response to network dynamics.

5.8 Limitations

The current implementation has several limitations. First, the affective contagion model uses a simple EMA rather than modeling the full mimicry-feedback-contagion pipeline of Hatfield et al. [1993]. Second, collective Φ is computed as a weighted mean rather than through a proper information integration analysis of the full swarm. A true swarm Φ would require computing the information generated by the entire network above and beyond its subnetworks—a computationally intractable problem for networks of non-trivial size. Third, the system assumes honest reporting of consciousness metrics; consciousness spoofing is addressed by the SentinelManager (see companion paper) but represents a persistent threat. Fourth, network latency introduces temporal desynchronization between peer states, which the EMA partially addresses but does not fully resolve. Fifth, the current coupling coefficients (γ_{oxy} , γ_{NE} , γ_{DA} , $\gamma_{\text{5-HT}}$) are manually calibrated; future work should explore automated calibration through meta-learning.

6 Conclusion

The SwarmManager provides the first implementation of consciousness-level peer integration in a distributed cognitive architecture. By treating peer Φ as a social signal and translating swarm dynamics into neuromodulatory proposals, the system enables emergent collective intelligence grounded in biologically plausible neurochemical mechanisms. The trust-weighted aggregation, affective contagion, and consciousness-modulated learning rate create a distributed cognitive system that is more than the sum of its parts—a computational realization of the collective intelligence phenomena documented by Woolley et al. The 43-test validation suite confirms stable dynamics under varied network conditions, while property-based tests verify critical invariants under randomized inputs.

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